

Repairability Is Not Enough: Product-Life Extension, Rebound-Risk Pathways and Enabling Conditions in Residential Window Systems

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Independent research project prepared as supporting material for research applications

Research focus:

Circular Building Components and Rebound Risks: Repair, Component Replacement, and Product-Life Extension in Residential Window Systems

Abstract

Circular design strategies often assume that maintenance, repair, component replacement and product-life extension reduce environmental impact by keeping products and materials in use for longer. In residential window systems, this assumption is plausible, but not automatic. A window may contain replaceable components, yet repair may still fail in practice if parts are inaccessible, spare parts are unavailable, installation details prevent future removal, or clients choose replacement for perceived improvement rather than technical necessity. This research therefore investigates the conditions under which repair and component replacement can extend the service life of residential window systems without their expected environmental benefits being weakened by rebound risks, implementation gaps, lifecycle trade-offs, burden shifting or technological lock-in.

The study uses a structured rapid literature review, qualitative evidence extraction, MAXQDA-based coding, one documented window-system case, two exploratory expert interviews, a qualitative lifecycle scenario comparison, seven classified risk pathways and a practitioner-reviewed preliminary screening framework. The selected case is the VELFAC 200 ENERGY top-guided residential window system, used as a publicly documented product case to examine component structure, repairability, uncertainty and life-extension potential.

The findings show that repairability is not only a product feature. It depends on practical and systemic conditions, including component access, installation method, spare-part availability, dimensional compatibility, supplier information, maintenance planning, actor responsibility and client decision-making. The most defensible conclusion is therefore conditional: component repair and replacement may support product-life extension, but only when the surrounding design, supply-chain and maintenance conditions make repair likely to happen in practice. The project contributes a preliminary framework, the Circular Window Component and Rebound-Risk Check, to help architects and specifiers screen these conditions before presenting circular window strategies as environmentally beneficial.

1. Introduction

Circular economy principles have become increasingly important in architecture and construction because the building sector is under pressure to reduce material waste, embodied environmental impacts and premature replacement of building components (Corona et al., 2019; Rose, 2019). In this context, repair, maintenance, component replacement, refurbishment and reuse are often presented as strategies that can extend product life and reduce the need for new production. However, these strategies do not automatically create environmental benefit simply because they sound circular. Their value depends on whether they are technically possible, practically used, supported by the supply chain and environmentally preferable over the full lifecycle.

Residential window systems are a useful case through which to examine this problem. Windows are not simple single-material products. They combine long-life elements, such as frames and sashes, with shorter-life or more failure-prone elements, such as hardware, hinges, handles, seals, gaskets, glazing units, coatings and weathering details. They also sit at an important interface between product design and building construction. Even when a window product includes repairable or replaceable parts, future repair may be limited by wall detailing, installation method, access, spare-part availability, certification, cost, client preference or lack of maintenance planning.

This makes windows a relevant architectural case for studying the gap between theoretical circularity and practical implementation. A strategy such as component replacement may appear beneficial because it avoids discarding the full window unit, but its benefit can be weakened if multiple service visits are required, if compatible parts are unavailable, if the retained frame prevents future performance upgrades, or if easier replacement encourages more frequent cosmetic or performance upgrading before functional failure (Kaddoura et al., 2019; Zink & Geyer, 2017; Castro et al., 2022). These possible outcomes should not all be described as rebound effects. Some are implementation gaps, some are lifecycle trade-offs, some are burden-shifting problems, and some are forms of technological lock-in. Distinguishing between these categories is central to the research.

The main research question is:

Under what design and system conditions can repair and component replacement extend the service life of residential window systems without their expected environmental benefits being substantially offset, and how can these findings inform a preliminary rebound-risk screening framework for architects and specifiers?

The project focuses on residential window systems in a primarily European context and uses a 50-year study period for the lifecycle scenario comparison. It does not attempt to produce a full lifecycle assessment or measure market-level rebound. Instead, it identifies plausible mechanisms, practical conditions and classification categories that can help architects and specifiers think more carefully before assuming that repairability or product-life extension is automatically environmentally beneficial.

2. Conceptual framework

This research is based on the idea that circular strategies should be assessed not only by their intended benefit, but also by the responses and conditions they create in practice. The central distinction is between rebound effects and other unintended or limiting outcomes. This distinction matters because not every failure of a circular strategy is a rebound effect. Some failures occur because the strategy cannot be implemented, while others occur because benefits are displaced to another lifecycle stage or reduced by additional material, transport or complexity.

In this project, a rebound effect is understood as a response generated or enabled by an intervention that offsets part of its expected environmental benefit (Zink & Geyer, 2017; Castro et al., 2022). In the context of window systems, a rebound-relevant example would be a situation where easier component replacement encourages more frequent upgrading before functional failure. The original intention may be to extend service life, but the behavioural response may increase the frequency of replacement events and reduce the expected benefit.

An implementation gap occurs when a circular strategy cannot function or is not used because necessary services, information, incentives, skills or infrastructure are missing. This is especially important for repairable window components. A handle, hinge or glazing unit may be technically replaceable, but if it cannot be accessed after installation, if spare parts are unavailable, or if no actor is responsible for maintenance, the repair strategy may fail in practice. In that case, the problem is not necessarily rebound. It is a gap between the technical possibility of repair and the practical conditions required to make repair happen.

A lifecycle trade-off occurs when improvement in one area creates additional impacts elsewhere (Corona et al., 2019; Kaddoura et al., 2019). A high-performance or more modular window may provide better operational performance or easier replacement of some components, but it may also require more materials, more complex mechanisms, additional connectors or more difficult maintenance. This does not mean that high-performance or modular systems are environmentally worse. It means that their benefit should be verified across the lifecycle rather than assumed from a single design feature.

Burden shifting occurs when environmental impact moves to another actor, lifecycle stage, location or impact category. For example, component repair may reduce whole-window replacement but increase service visits or transport. Full replacement may improve operational performance but create additional end-of-life burdens for the removed window. Direct reuse may avoid new production, but only if transport, adaptation, certification and storage do not eliminate the expected benefit.

Technological lock-in occurs when a long-lived solution restricts later environmental or technical improvement (Kaddoura et al., 2019; Mohamed, 2020). In window systems, retaining a durable frame can reduce material replacement, but it may also limit future glazing upgrades or performance improvements if the frame or product architecture cannot accommodate them. Long service life is therefore not automatically beneficial unless the retained system can continue to meet future functional and environmental requirements.

Finally, a general unintended consequence is an adverse outcome that does not necessarily offset the expected environmental benefit directly. For example, a repair-based strategy may be too difficult to communicate to clients, or too complex for typical maintenance routines, even if the environmental effect is unclear. This category is useful because it prevents the analysis from forcing every problem into the rebound category.

Together, these categories create the analytical structure of the project. They allow repair, component replacement and product-life extension to be examined as conditional strategies. The central question is not only whether a circular intervention looks beneficial on paper, but whether its expected benefit remains plausible once design details, actor responsibilities, maintenance conditions, user behaviour and lifecycle uncertainty are considered.

3. Methodology

This project used a structured exploratory research design combining a rapid literature review, qualitative coding, a product case study, expert interviews, a lifecycle scenario comparison, classified risk pathways and a preliminary practitioner-reviewed screening framework. The aim was not to produce a complete lifecycle assessment or statistically representative evidence, but to identify plausible conditions under which repair and component replacement may or may not lead to real product-life extension in residential window systems.

The research process was organised in six main stages. First, a rapid literature review was conducted to identify relevant academic and technical sources on circular building components, product-life extension, repairability, obsolescence, rebound effects, lifecycle trade-offs and implementation barriers. The search was organised into four search groups, and only sources available as full PDF documents were included in the final review. Each source was screened for relevance to the project's research question and then recorded in a literature search log, screening sheet and evidence matrix. These records are included in Appendix A.

Second, the selected literature was analysed using a predefined qualitative codebook in MAXQDA. The codebook was organised into five main groups: circular strategy, obsolescence, enabling condition, negative mechanism and mitigation. The circular strategy codes included maintenance, repair, component replacement, upgrade, refurbishment, direct reuse and recycling. The obsolescence codes captured physical, functional, technical, economic, regulatory, aesthetic and user-driven obsolescence. The enabling condition codes captured requirements such as component accessibility, disassembly, spare parts, repair information, skilled labour, service infrastructure, business models, regulation, certification and user acceptance. The negative mechanism codes captured possible risks such as premature replacement, increased material complexity, additional transport, burden shifting, lock-in, missing repair infrastructure and increased comfort or use. The mitigation codes captured design interventions, maintenance planning, supplier commitment, user guidance, business-model interventions, policy interventions and lifecycle verification. The exported

MAXQDA codebook, selected coded segments and code memos are included in Appendices B-D.

The purpose of the MAXQDA coding was to move from individual literature sources to recurring analytical patterns. Coding was therefore not treated as a separate end result, but as the analytical basis for the later case analysis, interviews, risk pathways and screening framework. The coded literature helped identify which circular strategies were repeatedly discussed, which barriers appeared across different sources, which conditions seemed necessary for implementation, and which unintended consequences should be distinguished from direct rebound effects.

Third, one real window-system case was selected to test the relevance of the literature findings in a concrete product context. The selected case was the VELFAC 200 ENERGY top-guided residential window system. This case was chosen because it is publicly documented and provides enough information to examine product composition, component structure, maintenance guidance, environmental documentation and uncertainty around repair and product-life extension. The case was not used to evaluate VELFAC as a company or to make a verified environmental claim about the product. Instead, it functioned as a structured example through which to examine the practical conditions that influence repairability and component replacement. The case documentation is included in Appendix E.

A component map was then created for the selected window system. This map separated long-life system components, such as the wood-aluminium frame system, exterior aluminium cladding, interior timber frame and main sash structure, from shorter-life or wear-prone components, such as hardware, handles, opening mechanisms, seals, gaskets, glazing-related elements, surface finishes and weathering details. The purpose of the component map was to identify where repair and component replacement could theoretically extend service life, and where practical risks might appear. The component map is included in Appendix F.

Fourth, two exploratory expert interviews were conducted. The first interview was with a window supplier and was used anonymously. The second interview was with an architect and was used with permission including name and organisation. The interviews were short and practical, focusing on common window failures, repairable components, spare-part availability, barriers to repair, client replacement decisions, reuse limitations, design decisions affecting repairability and information needs for architects. The same codebook used for the literature analysis was also used to interpret the interview responses. This allowed the interview findings to be connected directly to the literature themes, especially component accessibility, spare parts, implementation gaps, premature replacement, direct reuse barriers, lifecycle verification and design intervention. The interview log, question guide, notes, coding summary, consent statement and interview memo are included in Appendix G.

Fifth, a qualitative lifecycle scenario comparison was developed. The comparison used a 50-year study period and compared two main scenarios: full window replacement and component repair/replacement. A third possible scenario, refurbishment or direct reuse, was excluded from the main comparison because the evidence was not strong enough to model it robustly for the selected case. The lifecycle comparison did not calculate numerical environmental impacts. Instead, it compared the types of burden, uncertainty and enabling

conditions associated with each scenario. A separate assumptions table was created to make key assumptions transparent, including frame life, glazing life, hardware replacement intervals, repair events, transport distance, spare-part availability, component access and end-of-life route. The lifecycle scenario comparison is included in Appendix H, and the transparent assumptions are included in Appendix I.

Sixth, the findings were translated into seven classified risk pathways and a preliminary screening framework. The risk pathways used the structure: intervention, expected benefit, contextual response, possible environmental consequence, classification and mitigation. The classifications included implementation gap, potential behavioural rebound, lifecycle trade-off, burden shifting and technological lock-in. These pathways then informed the Circular Window Component and Rebound-Risk Check, a preliminary framework for architects and specifiers. The framework was organised around eight questions: function preservation, likely component failure, independent component replacement, practical repair enablement, actor responsibility, trade-offs, rebound or displacement risks and lifecycle plausibility. Finally, the framework was shown to one practitioner for feedback. This allowed the framework to be described as practitioner-reviewed, but not validated. The classified risk pathways are included in Appendix J, the preliminary screening framework is included in Appendix K, and the practitioner feedback and revisions are included in Appendix L.

The methodology has several limitations. The literature review was rapid rather than exhaustive. The MAXQDA coding was qualitative and interpretive. The product case relied on publicly available documentation. The interviews represented two practical perspectives and were not statistically representative. The lifecycle scenario comparison was qualitative and assumption-based rather than a complete LCA. The practitioner feedback came from one expert and therefore cannot validate the framework. These limitations are acknowledged throughout the report. However, the combination of literature coding, product analysis, interviews, scenario comparison and framework development provides a coherent exploratory method for identifying where circular window strategies may succeed, fail or require additional verification.

4. Literature findings

The coded literature did not support the idea that circular strategies are automatically beneficial simply because they extend the use of products or components. Instead, the findings showed that repair, maintenance, component replacement, reuse and recycling depend on specific technical, organisational and behavioural conditions. Across the coded sources, five main themes were especially relevant to residential window systems: circular strategies, obsolescence, enabling conditions, negative mechanisms and mitigation approaches.

4.1 Circular strategies: repair, maintenance and component replacement

The first recurring theme was the importance of distinguishing between different circular strategies. Repair, maintenance, component replacement, refurbishment, reuse and recycling

are often discussed together, but they do not work in the same way (Corona et al., 2019; Rose, 2019). Maintenance and repair aim to keep the existing product functioning. Component replacement aims to retain the main system while replacing shorter-life parts. Refurbishment usually involves a more substantial intervention. Direct reuse requires the product or component to be suitable for another context. Recycling usually occurs only after the product can no longer remain in use.

For residential window systems, component replacement is especially important because the window is made of parts with different expected lifespans (Mohamed, 2020; Búryová et al., 2025). A frame may remain structurally usable for a long time, while hardware, seals, glazing units, coatings or opening mechanisms may fail earlier. This supports the logic of retaining long-life components while maintaining or replacing shorter-life components. However, the literature also suggests that this strategy only works if those shorter-life components are identifiable, accessible, replaceable and supported by information, spare parts and service infrastructure.

A key finding from the literature is therefore that product-life extension should not be understood only as keeping the whole product in place for as long as possible. A more useful approach is to ask which parts need to last, which parts are expected to fail, and whether the design allows different component lifespans to be managed without forcing full-system replacement. This distinction became central to the later component map and screening framework.

4.2 Obsolescence: why products are replaced before physical failure

The second recurring theme was obsolescence. The coded literature showed that products and building components may become obsolete for several reasons, not only because they physically break (Litti et al., 2018; Femenella & Femenella, 2025). Physical obsolescence occurs when a component wears out or fails. Technical obsolescence occurs when a product no longer meets current performance expectations or available technical standards. Functional obsolescence occurs when the product no longer supports the required use. Economic obsolescence occurs when repair becomes more expensive or less attractive than replacement. Regulatory obsolescence occurs when requirements change. Aesthetic or user-driven obsolescence occurs when users prefer a newer, more modern or visually different solution before the existing product has reached technical end of life.

This is important for window systems because windows are both technical and visible building components. A client may replace a window because the glazing performance is poor, because the hardware no longer works, because the frame is damaged, because maintenance seems inconvenient, or because a newer and thicker product appears more desirable. These different reasons for replacement have different environmental implications. If a complete window is replaced because only a small component failed, the result may be premature replacement. If replacement occurs because the existing system no longer meets thermal or safety requirements, the decision may be more defensible, but still needs lifecycle verification.

The literature therefore supports a more careful definition of “failure.” A window system can fail physically, functionally, technically, economically or socially. A repair-based circular strategy must respond to these different forms of obsolescence. It is not enough to design a technically durable frame if user preference, poor maintenance access or future performance requirements still lead to early replacement.

4.3 Enabling conditions: why repairability depends on more than product design

The third recurring theme was the importance of enabling conditions. The coded literature showed that circular strategies depend on a network of supporting conditions (Droste, 2023; Grigoryan & Daskalakis, 2025; Järvelä et al., 2025). These include component accessibility, disassembly possibilities, spare-part availability, repair information, skilled labour, service infrastructure, business models, regulation, certification and user acceptance.

This finding is especially relevant to residential windows because a window is not only a product; it is also part of a building assembly. Even if a manufacturer designs replaceable components, repair may become difficult after the window is installed. Access can be blocked by finishes, reveals, sealants, wall build-up or detailing decisions. The ability to repair a hinge, handle, seal or glazing unit depends not only on the product, but also on the way the product is installed and maintained over time.

Spare-part availability also appeared as a key enabling condition (Mohamed, 2020; Kaddoura et al., 2019). Component replacement depends on compatible parts remaining available for a sufficient period. If parts become unavailable after a window system becomes older, a technically repairable product may still be replaced completely. Repair information is similarly important. Architects, clients and maintenance providers need to know what can be replaced, how replacement should happen, which parts are compatible, and what the expected maintenance schedule is.

The literature therefore suggests that repairability should be treated as a system condition. A product can be designed for repair, but repair also requires a service network, responsible actors, accessible components, clear documentation and a maintenance culture. This finding strongly influenced the project’s later distinction between technical repairability and practical repair implementation.

4.4 Negative mechanisms: rebound, implementation gaps, trade-offs, burden shifting and lock-in

The fourth recurring theme was the presence of negative mechanisms that can weaken the expected benefit of circular strategies. The literature supported the need to separate different types of negative outcomes instead of calling every problem a rebound effect (Zink & Geyer, 2017; Castro et al., 2022).

A rebound effect requires a response that offsets part of the expected environmental benefit of an intervention. In this project, the most relevant rebound pathway is behavioural: easier repair or replacement may create more opportunities for upgrading before functional failure. For example, if a user replaces visible or functional components more frequently because replacement is easier, the expected benefit of component-level repair may be reduced.

However, many problems identified in the literature are better described as implementation gaps. These occur when a circular strategy is technically possible but cannot be applied because practical conditions are missing. For example, a component may be replaceable in theory but inaccessible in practice. Spare parts may be unavailable. Skilled labour may be missing. Information may not be transferred from supplier to architect or from architect to client. In such cases, the problem is not necessarily rebound; it is the failure of the enabling system around the circular strategy.

Lifecycle trade-offs also appeared repeatedly, especially where durability, performance or circularity required additional materials, more complex assemblies or uncertain end-of-life routes (Carlisle & Friedlander, 2016; Búryová et al., 2025). A more durable, high-performance or modular system may provide benefits, but it may also require more material, more complex construction, more connectors, more labour or more difficult future maintenance. This means that circular design choices should be checked across the lifecycle. A strategy that reduces waste at one stage may add impacts at another stage.

Burden shifting was also relevant. Repair may reduce full-window replacement but increase service trips or transport. Full replacement may improve operational performance but create end-of-life burdens for the removed window. Reuse may avoid new production but require storage, transport, adaptation and certification. These shifts do not necessarily make the strategy wrong, but they must be made visible before environmental claims are made.

Technological lock-in was another important mechanism. A long-life component can reduce replacement, but it can also restrict future upgrades. For example, retaining a frame may be beneficial if it remains compatible with future glazing or performance requirements. If it prevents later improvement, however, the long-life strategy may create a future limitation. This finding is especially important for architects because design decisions made at installation can influence repair and upgrade possibilities many years later.

4.5 Mitigation: what makes circular strategies more credible

The fifth recurring theme was mitigation. The coded literature suggested that the risks described above can be reduced, but not by simply declaring a product circular. Mitigation requires concrete design, specification, maintenance and policy actions.

At the design level, components that are expected to fail earlier should be accessible, replaceable and documented. Detailing should allow future inspection, maintenance and removal where possible. Long-life components should be designed with future upgrade compatibility in mind, especially where performance standards may change over time.

At the supplier and service level, product-life extension depends on spare-part commitment, compatibility information, clear maintenance guidance and repair infrastructure. Without these, repairability may remain theoretical. Supplier information should not only describe product performance at the time of installation, but also support future decisions about maintenance, replacement, service life and end-of-life treatment.

At the project level, architects and specifiers need to ask for the right information early enough. The literature suggests that circular decisions become weaker when repair,

maintenance and end-of-life questions are left until after specification or installation. Early screening can help identify whether a circular strategy is practically enabled or whether it depends on assumptions that are not yet supported.

At the lifecycle level, mitigation requires verification through scenario comparison, lifecycle assessment, sensitivity analysis or other transparent methods of checking whether expected benefits remain plausible (Litti et al., 2018; Sällström Eriksson & Lidelöw, 2025). A repair or component-replacement strategy should be checked against service life, transport, repair frequency, replacement intervals, performance retention and end-of-life route. This does not always require a full LCA at early design stage, but it does require transparent assumptions and a clear statement of uncertainty.

Overall, the literature findings show that repair and component replacement can support product-life extension, but only under specific conditions. The strongest finding is that repairability is conditional. It depends on the relationship between product design, building integration, service infrastructure, actor responsibility, user behaviour and lifecycle impacts. This conclusion guided the case analysis, the expert interviews, the lifecycle scenario comparison, the classified risk pathways and the final screening framework.

5. Case analysis: VELFAC 200 ENERGY top-guided residential window

To connect the literature findings to a concrete product context, the study selected the VELFAC 200 ENERGY top-guided residential window system as a publicly documented case. The purpose of the case analysis was not to verify the environmental performance of this specific product or to evaluate the manufacturer. Instead, the case was used as a practical example through which to examine component structure, repairability, documentation, uncertainty and possible product-life extension conditions. The detailed case documentation is provided in Appendix E, and the component map is provided in Appendix F.

The selected window system is relevant because it combines several features that are common in contemporary residential window design: a wood-aluminium construction, triple glazing, factory-finished components, opening hardware, seals, weathering details and manufacturer documentation (Búryová et al., 2025; Carlisle & Friedlander, 2016). This makes it suitable for analysing the difference between long-life system components and shorter-life wear components. It also makes the case useful for examining whether component replacement can realistically support service-life extension without requiring full-window replacement.

5.1 Component structure and service-life logic

The component map created for the case separated the window into long-life system components, short-life or wear-prone components, repairable or replaceable components, and risk points. The long-life components include the wood-aluminium frame system, exterior aluminium cladding, interior timber frame, main sash structure, opening geometry and wall/window interface. These components form the main product structure and are the parts that would ideally be retained if a repair or component-replacement strategy is successful.

The shorter-life or wear-prone components include the opening hardware, hinges or friction arms, handles, locking elements, seals, gaskets, glazing unit edge seals, surface finish, drainage details and other weather-exposed or moving parts. These components are more likely to require inspection, adjustment, repair or replacement before the entire window reaches the end of its service life. This supports the central logic of the project: if shorter-life components can be replaced while long-life components remain in use, then product-life extension may be possible.

However, the case also shows why this logic cannot be assumed. A component being theoretically replaceable does not mean that it will be replaced in practice. Repair depends on whether the component can be accessed after installation, whether compatible replacement parts exist, whether the repair sequence is documented, whether the cost and effort are acceptable, and whether the client chooses repair instead of full replacement. The component map therefore does not prove that the system is circular. It identifies where repairability may be possible and where practical barriers may appear.

5.2 Repairability and documentation

The VELFAC case includes product documentation, environmental documentation and maintenance-related guidance. This is important because repair and product-life extension depend partly on information availability. A window system that has no accessible documentation is harder to evaluate, specify, maintain or repair. In this case, the existence of documentation supports a more structured analysis of the system.

At the same time, the available information does not fully answer all circularity questions. Public documentation can show product composition, performance claims, maintenance instructions or selected replacement guidance, but it does not always provide enough detail about long-term spare-part availability, actual repair frequency, component-level environmental impacts, transport, end-of-life recovery, reuse routes or the consequences of different installation methods. These uncertainties are important because they affect whether repair and component replacement can be treated as reliable product-life extension strategies.

The case therefore highlights a recurring gap between product documentation and lifecycle decision-making. Architects and specifiers may receive enough information to select and install a window, but not always enough information to judge whether the system will remain repairable, upgradeable or environmentally beneficial over several decades. For circular design, the most important information is not only initial performance, but also what happens after years of use: which components fail, which parts can be replaced, how replacement happens, who provides parts, and what happens at the end of life.

5.3 Installation and the wall/window interface

One of the strongest findings from the case analysis and interviews is that repairability is not contained only inside the product. It is also shaped by the way the window is installed into the building. The wall/window interface can either support future repair or make it difficult. If hinges, mechanisms, seals or glazing-related elements are blocked by finishes, reveals,

sealants or inaccessible detailing, repair may become impractical even when the product itself contains replaceable parts.

This is especially important for architects because the decision is not only about choosing a window product. It is also about specifying details that allow future access, inspection, maintenance, adjustment and replacement. A repairable product can become difficult to repair if it is embedded in a way that prevents access. In that case, the problem is best understood as an implementation gap: the strategy exists technically, but the surrounding project conditions prevent it from functioning.

The case therefore supports a more architectural definition of repairability. Repairability is not only whether a manufacturer can replace a part in theory. It is whether the product, installation detail, maintenance process and responsible actors together allow that replacement to happen in practice.

5.4 Reuse and end-of-life uncertainty

Direct reuse was considered but excluded from the main lifecycle scenario comparison because the evidence was not strong enough to model it robustly for this case, and reuse of building components often depends on dimensional compatibility, documentation, certification, logistics and market demand (Rose, 2019; Yang et al., 2024). The interviews supported this decision. Both expert perspectives indicated that direct reuse of removed windows is difficult, especially where windows are made to measure for specific openings. Even if a window can be removed, reuse may be limited by dimensions, ageing components, certification uncertainty, transport, storage and lack of local reuse infrastructure.

This does not mean that reuse is impossible. It means that direct reuse should not be assumed as a reliable pathway without stronger evidence. In the selected case, retaining the existing frame and replacing shorter-life components is a more defensible life-extension strategy than assuming that the entire removed window could easily be reused elsewhere.

End-of-life treatment also remains uncertain. Recycling may be possible for some materials, especially aluminium and glass, but the actual recovery route depends on local systems, separation quality, material condition and available infrastructure. This creates a risk of burden shifting. A strategy may appear beneficial during use but still create environmental burdens at removal, disposal, recycling or downcycling stages. For this reason, the lifecycle scenario comparison treats end-of-life as an uncertainty rather than a fixed benefit.

5.5 Case analysis conclusion

The VELFAC case supports the overall finding that repair and component replacement can be plausible product-life extension strategies, but only under specific conditions. The case contains long-life components that could theoretically be retained and shorter-life components that may require maintenance or replacement. However, the environmental value of this strategy depends on access, installation method, compatible spare parts, maintenance planning, actor responsibility, performance retention and end-of-life handling.

The case therefore does not lead to a simple conclusion that component replacement is always better than full replacement. Instead, it shows why a screening framework is needed. Architects and specifiers need a way to ask whether repair is technically possible, practically enabled and environmentally plausible before presenting a window strategy as circular or sustainable.

6. Lifecycle scenario comparison and classified risk pathways

The lifecycle scenario comparison was developed to examine the conditions under which product-life extension may remain environmentally plausible, following the broader literature finding that longer product life is not automatically beneficial without lifecycle verification (Kaddoura et al., 2019; Sällström Eriksson & Lidelöw, 2025). It was not designed as a complete lifecycle assessment. No exact environmental-impact values were calculated because reliable product-specific data for repair events, transport, use-phase changes and end-of-life treatment were not available. Instead, the comparison used transparent assumptions to identify where each strategy may create benefits, uncertainties or risks. The lifecycle scenario comparison is documented in Appendix H, and the transparent assumptions and uncertainty notes are documented in Appendix I.

The comparison used a 50-year study period and one residential window opening as the functional unit. The compared function included equivalent thermal, acoustic, safety, weather-protection and operational performance. This was important because repair and replacement can only be compared fairly if they provide the same required function. If a repaired window no longer meets performance requirements, then repair cannot automatically be treated as equivalent. If a replacement window improves performance significantly, that benefit must be considered, but it must also be weighed against additional production, transport, installation and end-of-life burdens.

6.1 Compared scenarios

Two main scenarios were compared. Scenario A represented full window replacement. In this scenario, the existing window is fully replaced at a chosen interval within the 50-year study period. This includes the production of a replacement window, transport, installation, removal of the old window and end-of-life treatment. The intended benefit of this scenario is that a new window system may improve performance, durability or user satisfaction. The main uncertainty is whether the operational or functional improvement is large enough to justify the additional material, production and end-of-life burdens.

Scenario B represented component repair and replacement. In this scenario, the long-life frame and main system structure are retained while shorter-life components are maintained, adjusted or replaced when needed. This may include hardware replacement, handle or fitting replacement, seal or gasket replacement, glazing unit replacement, inspection, cleaning and service visits. The intended benefit of this scenario is to extend the life of the whole system by avoiding premature full-window replacement. The main uncertainty is whether repair is

practically enabled over time through access, spare parts, skilled labour, maintenance planning and performance retention.

A third scenario, refurbishment or direct reuse, was considered but excluded from the main comparison. This decision was made because direct reuse was not sufficiently supported by the case evidence or interviews. Removed windows may be difficult to reuse because of custom dimensions, certification uncertainty, ageing components, transport and lack of matching demand. For this reason, direct reuse was treated as a future research option rather than a robust scenario in this study.

6.2 Transparent assumptions

The comparison used transparent assumptions instead of fixed environmental-impact values. The main assumptions concerned frame life, glazing life, hardware replacement interval, seal or gasket replacement interval, full replacement timing, number of repair events, transport distance, service visit grouping, spare-part availability, component access, end-of-life route and direct reuse potential.

The baseline assumption treated the study period as 50 years. The frame was treated as a long-life component that may plausibly remain in use for several decades if condition, performance and maintenance are adequate. Glazing, hardware, seals and fittings were treated as shorter-life components that may require replacement or service before the frame reaches end of life. Full replacement in Scenario A was placed at a plausible mid-life point for comparison, while Scenario B included multiple smaller interventions over time.

The transparent assumptions table was important because it prevented false precision. Without reliable data, it would be misleading to claim exact environmental savings from repair or replacement. Instead, the assumptions made clear which conditions would make each scenario stronger or weaker. If the frame lasts longer and components remain accessible, Scenario B becomes more plausible. If spare parts are unavailable, repair events are frequent, transport distances are long or performance cannot be maintained, the benefit of Scenario B becomes less certain. If replacement produces a large verified operational improvement, Scenario A may become more defensible, but only if the added embodied and end-of-life burdens are also considered.

6.3 Interpretation of the lifecycle comparison

The comparison suggests that component repair and replacement are most likely to be beneficial when the main frame remains functional, shorter-life parts can be replaced independently, repair visits are not excessive, compatible spare parts remain available and the repaired system continues to meet performance requirements. Under these conditions, Scenario B may reduce whole-window material demand and delay end-of-life burdens.

However, Scenario B is vulnerable to implementation gaps. If components are inaccessible, if parts cannot be sourced, if the installation detail prevents service, or if no actor is responsible for maintenance, the strategy may fail and lead to full replacement anyway. This means that the environmental benefit of repair depends not only on the product design but also on the maintenance and supply-chain system around it.

Scenario A may be more defensible when the existing window performs poorly, when the frame or main structure is no longer serviceable, or when replacement produces a large verified improvement in operational performance (Litti et al., 2018; Sällström Eriksson & Lidelöw, 2025). However, full replacement also creates a complete new product burden and an end-of-life burden for the removed window. It may also become premature replacement if chosen because of preference for a newer or more modern product rather than functional need.

The strongest conclusion from the lifecycle comparison is therefore not that one scenario is always better. The stronger conclusion is that each scenario depends on conditions that must be checked before making environmental claims. Repair is not automatically beneficial. Replacement is not automatically wasteful. The correct decision depends on service life, performance, access, parts, transport, user behaviour and end-of-life route.

6.4 Classified risk pathways

The lifecycle comparison and earlier findings were translated into seven classified risk pathways. These pathways were used to show how an intended circular intervention may produce benefits, but also how those benefits may be weakened by contextual responses or missing conditions. The full pathway table, report-ready pathway chains and synthesis notes are included in Appendix J.

The first pathway concerned technically repairable hardware, hinges, handles or fittings. The expected benefit is longer service life through replacement of short-life components. The contextual risk is that physical access may be blocked by finishes, reveals, the wall/window interface or installation detailing. The possible consequence is full-window replacement even though the failed part is technically repairable. This was classified as an implementation gap and is one of the strongest pathways in the study.

The second pathway concerned replaceable components and spare parts. The expected benefit is future product-life extension through repair. The contextual risk is that compatible spare parts may become unavailable or difficult to source as the window system becomes older. The possible consequence is complete window replacement earlier than technically necessary. This was also classified as an implementation gap.

The third pathway concerned replaceable visible or functional components. The expected benefit is easier maintenance, adjustment and targeted upgrading. The contextual risk is that repair opportunity becomes an opportunity for cosmetic or performance upgrading before functional failure. The possible consequence is more frequent replacement of components or whole windows. This was classified cautiously as a potential behavioural rebound pathway, not as a measured rebound effect.

The fourth pathway concerned highly modular or high-performance window systems. The expected benefit is easier disassembly, better performance and possible component replacement. The contextual risk is that added connectors, materials, mechanisms or technical complexity may be introduced to enable modularity or higher performance. The possible consequence is that higher initial production burden may reduce or offset part of the expected lifecycle benefit. This was classified as a lifecycle trade-off.

The fifth pathway concerned retaining a long-life frame. The expected benefit is reduced need for full-window replacement and lower whole-system material demand. The contextual risk is that future glazing or performance upgrades may be restricted by the retained frame, interface or product architecture. The possible consequence is extended use of a technically outdated system or disruptive replacement later. This was classified as technological lock-in.

The sixth pathway concerned component repair and maintenance strategy. The expected benefit is lower material demand compared with full replacement. The contextual risk is that multiple service visits, specialist labour or long-distance transport may be needed over the study period. The possible consequence is that expected material savings are partly reduced by transport, labour or repeated intervention burdens. This was classified as burden shifting.

The seventh pathway concerned reuse of removed full windows or major components. The expected benefit is avoided new production through direct reuse or refurbishment. The contextual risk is that project-specific dimensions, certification uncertainty, ageing components and transport make reuse difficult. The possible consequence is that reuse potential is not realised or that additional adaptation and transport burdens reduce the expected benefit. This was classified as an implementation gap or burden shifting.

6.5 Lifecycle and pathway conclusion

The lifecycle scenario comparison and risk pathways show that the environmental value of repair and component replacement is conditional. The most important risks are not only technical failure, but practical and systemic weaknesses: blocked access, unavailable parts, unclear responsibilities, transport burdens, client preference, uncertain end-of-life routes and future upgrade limitations.

The classified pathways also show why it is important not to call every negative outcome a rebound effect. Only one pathway in this study is strongly rebound-relevant: the possibility that easier replacement encourages more frequent upgrading before failure. Other pathways are better described as implementation gaps, lifecycle trade-offs, burden shifting or technological lock-in. This distinction makes the analysis more precise and helps architects respond with the right mitigation strategy.

Overall, the lifecycle and pathway analysis supports the need for a preliminary screening framework. Before a repair or component-replacement strategy is described as circular, architects and specifiers should check whether the strategy preserves the required function, whether short-life components can be replaced independently, whether repair is practically enabled, which actors must support it, what trade-offs it introduces, whether rebound or displacement may occur, and whether the expected lifecycle benefit remains plausible under transparent assumptions.

7. Preliminary screening framework: Circular Window Component and Rebound-Risk Check

The final output of the project is a preliminary screening framework called the Circular Window Component and Rebound-Risk Check. The framework translates the literature findings, VELFAC case analysis, expert interviews, lifecycle scenario comparison and classified risk pathways into a practical tool for architects and specifiers, building on the need for practical circularity and disassembly assessment methods in design decision-making (Droste, 2023; Järvelä et al., 2025). Its purpose is not to calculate environmental impact values or replace a complete lifecycle assessment. Instead, it helps identify whether a repair, component-replacement or product-life extension strategy is likely to be practically enabled, and whether its expected environmental benefit may be weakened by rebound-related behaviour, implementation gaps, lifecycle trade-offs, burden shifting or technological lock-in. The framework documents are included in Appendix K.

The framework was developed because the research showed that repairability cannot be judged only from the presence of replaceable parts. A window component may be technically replaceable, but repair may still fail if access is blocked, spare parts are unavailable, installation detailing prevents future removal, actors do not take responsibility, or clients prefer replacement before technical failure. The framework therefore checks both product-level and system-level conditions.

7.1 Structure of the framework

The framework is organised around eight screening questions.

The first question asks what function the window system must preserve over the study period. This is necessary because repair and replacement strategies can only be compared fairly if they deliver equivalent performance. The required function includes thermal performance, acoustic performance, safety, weather protection, operation, durability and user comfort. Without a clear functional unit, a repair strategy may be unfairly compared with a replacement strategy that provides a different performance level.

The second question asks which components are likely to fail before the whole window reaches end of life. This question separates long-life system components from shorter-life wear components. In the selected case, likely shorter-life components include hardware, opening mechanisms, handles, fittings, seals, gaskets, glazing unit edge seals and surface finishes. This question is important because full replacement may be chosen even when only a smaller component has failed.

The third question asks whether short-life components can be replaced independently without replacing the complete window. This tests technical repairability. It requires evidence such as removable parts, replaceable glazing units, available fittings, compatibility information and documented repair sequences. If independent replacement is not confirmed, product-life extension remains theoretical.

The fourth question asks whether repair is practically enabled by access, installation method and maintenance infrastructure. This question responds directly to one of the strongest findings of the project: repair can fail even when it is technically possible. Practical repair depends on physical access, service clearance, wall/window detailing, skilled labour, maintenance guidance and local service availability. This question is especially relevant for architects because installation decisions made at design and construction stages can determine whether future repair is possible.

The fifth question asks which actors must support the strategy over time. Repair and component replacement depend on manufacturers, suppliers, installers, architects, clients, maintenance providers, building managers, recyclers and possible reuse actors. If responsibility is unclear, a technically repairable system may still fail in practice. This question therefore turns repairability into an actor and responsibility issue, not only a product issue.

The sixth question asks what trade-offs are introduced by the circular strategy. A system that is more durable, modular or high-performing may also introduce added material, technical complexity, cost, labour, transport or future upgrade limitations. This question prevents circular design strategies from being treated as automatically beneficial. It connects the framework directly to the lifecycle scenario comparison.

The seventh question asks whether the strategy could trigger rebound, displacement or unintended replacement behaviour. This question checks whether the intervention may change behaviour or shift environmental burdens elsewhere. For example, easier component replacement may encourage premature upgrading, while repair strategies may shift impacts to service visits or transport. This question is important because it separates actual rebound-relevant risks from ordinary implementation problems.

The eighth question asks whether the expected lifecycle benefit remains plausible after assumptions and uncertainties are considered. This final question connects the screening framework to the transparent assumptions table. It requires the decision-maker to consider service life, repair frequency, spare-part availability, component access, transport distance, end-of-life route and operational performance. If the expected benefit depends on uncertain or unsupported assumptions, the framework does not reject the strategy, but it requires the claim to be presented as conditional.

7.2 Trial application to the VELFAC case

The framework was trialled on the VELFAC 200 ENERGY top-guided residential window case. This trial application showed that the framework can identify both the potential and the uncertainty of a repair-based strategy.

For the first question, the functional requirement was defined as one residential window opening providing equivalent thermal, acoustic, safety, weather-protection and operational performance over 50 years. This made it clear that repair and replacement can only be compared if they preserve the same basic function.

For the second question, the trial identified several shorter-life components that may fail before the main frame, including hardware, handles, fittings, seals, gaskets, glazing-related components and surface finishes. This supported the component-replacement logic of retaining long-life elements while addressing shorter-life elements.

For the third question, the trial found that some components appear repairable or replaceable, including selected fittings, handles, hardware and possibly glazing-related components. However, public information was not sufficient to fully confirm every replacement route, long-term compatibility or repair sequence. The framework therefore classified technical repairability as partly confirmed rather than fully proven.

For the fourth question, the trial showed that practical repair depends strongly on installation access, service clearance, the wall/window interface and maintenance infrastructure. This was one of the most important findings because it showed that product documentation alone is not enough. A technically repairable window may become difficult to repair if the building detail prevents access.

For the fifth question, the trial identified multiple actors who must support repair over time: manufacturer, supplier, installer, architect, client and maintenance provider. This reinforced the idea that product-life extension is a shared responsibility across the supply chain.

For the sixth question, the trial showed that the wood-aluminium frame and triple glazing may support durability and performance, but may also involve material and complexity trade-offs that require lifecycle verification. The framework therefore avoided treating durability or high performance as automatically beneficial.

For the seventh question, the trial identified possible rebound or displacement risks, including client preference for newer products, repeated service transport, difficult direct reuse due to custom dimensions and uncertain end-of-life recovery. These risks were not forced into one category. Instead, they were classified separately as potential behavioural rebound, burden shifting, implementation gap or lifecycle uncertainty.

For the eighth question, the trial concluded that component repair and replacement may be a plausible life-extension strategy for the selected window system if access, spare parts, maintenance planning and performance retention are ensured. However, the environmental benefit cannot be claimed quantitatively without stronger data. The most appropriate conclusion is therefore conditional rather than definitive.

7.3 Practitioner feedback

The framework was shown to one practitioner who had also participated in the expert interviews. The purpose of this feedback round was to check whether the framework was understandable, whether it could be used in architectural practice, which questions might be unrealistic, what information an architect would realistically have, what was missing, and whether the framework distinguished rebound-related risks from ordinary implementation problems. The feedback materials, response log and revision table are included in Appendix L.

The practitioner stated that the framework was understandable. He indicated that it could be used at any project stage, although it would be best used at the beginning of a project and could also be applied to an already started project. He did not identify any question as unrealistic or difficult to answer. He stated that an architect would have enough information to use the framework, did not identify missing elements, and confirmed that the framework distinguishes rebound risks from ordinary implementation problems.

This feedback did not require structural changes to the framework. The main revision was to clarify its practical positioning. The framework is best used early in the project, when product selection, specification and detailing decisions can still be influenced. However, it may also be used later as a review tool for already-started projects. Because the feedback came from one practitioner only, the framework can be described as practitioner-reviewed, but not validated.

7.4 Framework contribution

The main contribution of the framework is that it turns circular design from a general intention into a set of checkable conditions. Instead of asking only whether a window is durable, repairable or high-performing, the framework asks whether the required function is preserved, which parts fail first, whether those parts can be replaced independently, whether repair is practically enabled, which actors must support the strategy, what trade-offs are introduced, whether rebound or displacement may occur, and whether the lifecycle benefit remains plausible.

This is useful for architects and specifiers because many circular decisions are made before full evidence is available. At early design stage, it may not be possible to conduct a complete lifecycle assessment. However, it is still possible to ask better questions, identify weak assumptions and avoid unsupported sustainability claims. The framework therefore functions as an early warning and decision-support tool. It does not prove environmental benefit, but it helps clarify what must be true for the benefit to be credible.

8. Limitations

This project has several limitations that must be stated clearly. First, the literature review was rapid and structured, but not exhaustive. The review focused on sources available as PDFs within the project timeframe. This means that relevant literature may have been excluded if it was unavailable, inaccessible or outside the selected search groups. The findings should therefore be understood as exploratory rather than comprehensive.

Second, the MAXQDA coding was qualitative and interpretive. The codebook provided structure and helped identify recurring themes, but coding decisions still involved judgement. The coding should therefore be understood as a transparent analytical method, not as a statistically objective measurement of the literature.

Third, the case analysis relied on publicly available documentation for one window system. The VELFAC 200 ENERGY case provided enough information to examine product structure,

documentation and uncertainty, but it did not provide complete data on real repair frequency, long-term spare-part availability, project-specific installation conditions, transport, actual end-of-life routes or quantified environmental impacts. The case should therefore be understood as a practical example, not as a verified product assessment.

Fourth, the expert interviews were exploratory and limited to two perspectives: one window supplier and one architect. These interviews provided useful practical insight, especially on component failure, repair barriers, spare parts, client preferences, installation methods and standardisation. However, they are not representative of the wider market. More interviews with manufacturers, installers, maintenance providers, recyclers, clients and public-sector actors would be needed to strengthen the findings.

Fifth, the lifecycle scenario comparison was qualitative and assumption-based. It did not calculate embodied carbon, operational energy savings, transport impacts or end-of-life impacts. This was a deliberate choice because reliable product-specific data for all relevant variables were not available. The comparison therefore identifies plausible trade-offs and uncertainties rather than exact environmental savings.

Sixth, the rebound-risk pathways identify plausible mechanisms, not measured effects. The project does not prove that rebound effects are occurring in the window market. It identifies where rebound-relevant behaviour could appear, and where other classifications such as implementation gap, lifecycle trade-off, burden shifting or technological lock-in are more accurate.

Finally, the screening framework was reviewed by one practitioner only. This improves its practical relevance, but does not validate it. A validated framework would require repeated testing across multiple projects, practitioners, product types, countries and decision-making contexts.

These limitations do not undermine the value of the project, but they define its claim level. The research should be read as a preliminary, exploratory and practice-oriented study. Its strength is not in producing final environmental impact numbers, but in clarifying the conditions that must be checked before circular window strategies are presented as environmentally beneficial.

9. Conclusion

This research investigated the conditions under which repair and component replacement can extend the service life of residential window systems without their expected environmental benefits being weakened by rebound risks, implementation gaps, lifecycle trade-offs, burden shifting or technological lock-in. The project began from a simple but important concern: circular strategies are often presented as beneficial because they keep products and materials in use for longer, but this benefit is not automatic once design, installation, maintenance, user behaviour and lifecycle uncertainty are considered.

The literature findings showed that repair, maintenance, component replacement, reuse and recycling depend on specific enabling conditions. They also showed that products may be

replaced for many reasons, including physical failure, technical obsolescence, economic decisions, regulatory change, aesthetic preference and user-driven expectations. This means that product-life extension cannot be assessed only by asking whether the product is durable. It must also ask why replacement happens, which components fail, and whether smaller interventions can realistically avoid full-system replacement.

The case analysis of the VELFAC 200 ENERGY top-guided residential window showed that component repair and replacement may be plausible, but only conditionally. The system includes long-life components that could theoretically be retained and shorter-life components that may require maintenance or replacement. However, the practical value of this strategy depends on component access, installation method, spare-part availability, maintenance guidance, actor responsibility, dimensional compatibility, performance retention and end-of-life route.

The expert interviews reinforced this conclusion. Both experts confirmed that components such as mechanisms, handles, fittings and glazing units may fail or require replacement before the full window system. They also confirmed that repairability depends strongly on access, installation and practical decision-making. The supplier perspective emphasised common failures, spare parts and client preference for newer products. The architect perspective emphasised installation method, dimensional standardisation, supplier information, client briefs and the role of early design decisions. Together, the interviews supported the project's distinction between technical repairability and practical implementation.

The lifecycle scenario comparison showed that neither full replacement nor component repair is automatically better. Full replacement may be justified if the existing window no longer performs adequately or if replacement produces a verified operational improvement. However, it also adds production, transport, installation and end-of-life burdens. Component repair and replacement may reduce whole-system material demand and delay full-window disposal, but only if repair is accessible, spare parts remain available, service visits are reasonable and the repaired window continues to meet performance requirements.

The classified risk pathways showed that different negative outcomes require different classifications. Blocked access and unavailable spare parts are best understood as implementation gaps. Added material or technical complexity is a lifecycle trade-off. Repeated service transport or uncertain end-of-life treatment can create burden shifting. Retaining a long-life frame that restricts future upgrades can create technological lock-in. Easier replacement encouraging premature upgrading is the most rebound-relevant pathway, but it should be presented cautiously as a potential behavioural rebound mechanism, not as a measured rebound effect.

The final framework, the Circular Window Component and Rebound-Risk Check, brings these findings together into a practical screening tool. It helps architects and specifiers ask eight questions before treating a repair or component-replacement strategy as environmentally beneficial. These questions cover function, component failure, independent replacement, practical repair conditions, actor responsibility, trade-offs, rebound or displacement, and lifecycle plausibility. Practitioner feedback confirmed that the framework

was understandable and usable in architectural practice, especially early in the project, while also remaining applicable to already-started projects.

The main conclusion is that repairability is necessary, but not sufficient. A window system may be technically repairable, but product-life extension only becomes environmentally credible when repair is practically enabled and lifecycle assumptions remain plausible.

Circular design therefore needs to move beyond product claims and include the surrounding system: installation details, maintenance infrastructure, spare-part supply, actor responsibility, user behaviour and end-of-life routes.

For architects, the implication is practical. The most important circular design question is not only “Can this product last longer?” but also “What has to be true for this longer life to actually happen, and what unintended consequences could weaken the expected benefit?”

Asking this question early can help prevent circular strategies from becoming only theoretical. It can also support more honest, evidence-based decisions about repair, replacement and product-life extension in residential window systems.

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Appendices

The appendices are submitted as separate supporting PDF files. They provide the audit trail for the rapid literature review, qualitative coding, case documentation, interviews, lifecycle scenario comparison, transparent assumptions, classified risk pathways, preliminary screening framework and practitioner feedback.

Appendix A - Literature Search and Evidence Records

This appendix includes the completed literature search records and evidence matrix used in the rapid literature review.

Included files:

- A1a_Search_Log.pdf
- A1b_Candidate_Sources.pdf
- A1c_Full_Text_Screen.pdf
- A2_Evidence_Matrix.pdf

These records show how sources were identified, screened and selected for analysis. They document the search groups, candidate sources, full-text screening decisions and extracted evidence used to support the literature findings.

Appendix B - MAXQDA Codebook

Included file:

- B_MAXQDA_Codebook.pdf

This appendix includes the qualitative codebook used to analyse the selected literature. The codebook is organised into five main groups: circular strategy, obsolescence, enabling condition, negative mechanism and mitigation. It documents the coding structure used to identify recurring analytical patterns.

Appendix C - Selected Coded Literature Segments

Included file:

- C_MAXQDA_Selected_Coded_Segments.pdf

This appendix includes selected coded literature segments exported from MAXQDA. The file provides an audit trail for the qualitative coding process by showing coded passages, source information, assigned codes and overlapping code applications.

Appendix D - MAXQDA Code Memos and Coding Notes

Included file:

- D_MAXQDA_Code_Memos_and_Coding_Notes.pdf

This appendix includes the MAXQDA code memo export and coding notes used during the literature analysis. These materials document how codes were defined and applied during the coding process.

Appendix E - Case Documentation

Included file:

E_Case_Documentation.pdf

This appendix includes the VELFAC 200 ENERGY case documentation table. It records the selected case, available product information, component details, maintenance and replacement information, environmental documentation, repairability evidence and key uncertainties.

Appendix F - Component Map

Included file:

F_VELFAC_200_Component_Map.pdf

This appendix includes the component map for the VELFAC 200 ENERGY top-guided residential window system. The map separates long-life system components, shorter-life wear components, repairable or replaceable parts, risk points, full-system replacement risk and lifecycle verification needs.

Appendix G - Expert Interview Materials

Included files:

G1a_Interview_Log.pdf

G1b_Interview_Notes.pdf

G1c_Interview_Coding_Summary.pdf

G2_Interview_Consent.pdf

G3_Interview_Question_Guide.pdf

G4_INT01_Window_Supplier_Notes.pdf

G5_INT02_Architect_Notes.pdf

G6_Interview_Memo.pdf

This appendix includes the interview consent and use-of-notes statement, interview question guide, interview log, structured interview notes, interview coding summary and interview memo. These documents show how the exploratory practitioner evidence was collected, interpreted and connected to the literature coding, case analysis and risk pathways.

Appendix H - Lifecycle Scenario Comparison

Included files:

H1_Functional_Unit.pdf

H2_Scenario_Definitions.pdf

H3_50_Year_Timeline.pdf

H4_Qualitative_Comparison.pdf

This appendix includes the qualitative lifecycle scenario comparison. It documents the functional unit, scenario definitions, 50-year timeline and qualitative comparison between full window replacement and component repair or replacement.

Appendix I - Transparent Assumptions

Included files:

I1_Assumptions_Table.pdf
I2_Uncertainty_Notes.pdf
I3_Scenario_Implications.pdf

This appendix includes the transparent assumptions table, uncertainty notes and scenario implications. These records explain the assumptions used for frame life, glazing life, hardware replacement, repair events, transport, spare-part availability, component access, end-of-life route and environmental-impact uncertainty.

Appendix J - Classified Risk Pathways

Included files:

J1_Pathway_Table.pdf
J2_Report_Ready_Pathways.pdf
J3_Synthesis_and_Limitations.pdf

This appendix includes the seven classified risk pathways developed from the literature findings, case analysis, interviews and lifecycle scenario comparison. Each pathway follows the structure: intervention, expected benefit, contextual response, possible environmental consequence, classification and mitigation.

Appendix K - Preliminary Screening Framework

Included files:

K1_Framework_Questions.pdf
K2_Checklist_Template.pdf
K3_VELFAC_Trial_Application.pdf
K4_Risk_Classification.pdf
K5_Report_Ready_Text.pdf

This appendix includes the Circular Window Component and Rebound-Risk Check, including the framework questions, checklist template, VELFAC trial application, risk classification guide and report-ready framework text.

Appendix L - Practitioner Feedback and Revisions

Included files:

L1_Feedback_Questions.pdf
L2_Practitioner_Response_Log.pdf
L3_Revision_Table.pdf
L4_Report_Ready_Text.pdf

This appendix includes the practitioner feedback questions, response log, revision table and final report-ready wording. It documents the feedback round used to describe the framework as practitioner-reviewed, but not validated.